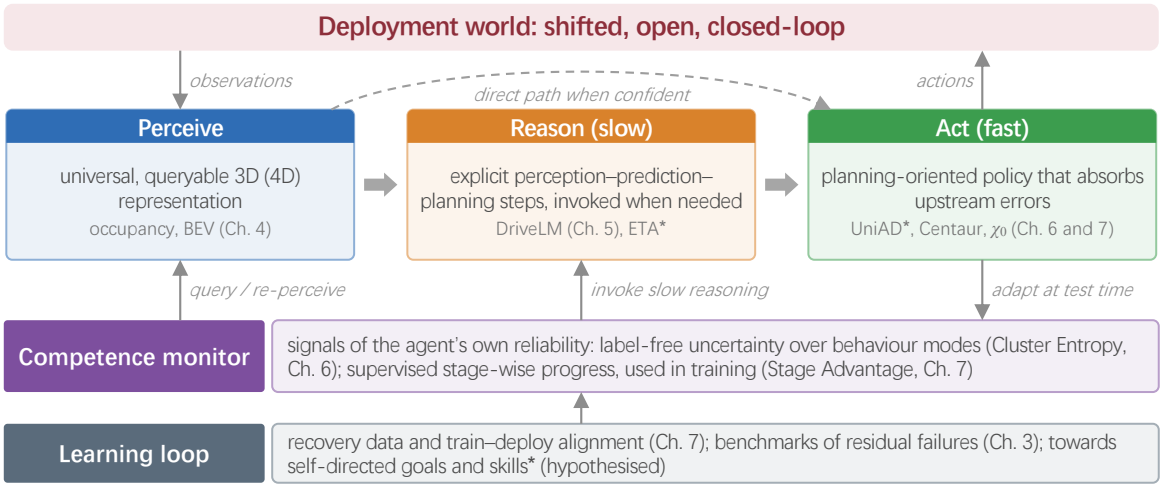

Deployment world: shifted, open, closed-loop
observations
direct path when confident
actions
Perceive
universal, queryable 3D (4D) representation
occupancy, BEV (Ch. 4)
Reason (slow)
explicit perception–prediction–planning steps, invoked when needed
DriveLM (Ch. 5), ETA*
Act (fast)
planning-oriented policy that absorbs upstream errors
UniAD*, Centaur, χ_0 (Ch. 6 and 7)
query / re-perceive
invoke slow reasoning
adapt at test time
Competence monitor
signals of the agent's own reliability: label-free uncertainty over behaviour modes (Cluster Entropy, Ch. 6); supervised stage-wise progress, used in training (Stage Advantage, Ch. 7)
Learning loop
recovery data and train–deploy alignment (Ch. 7); benchmarks of residual failures (Ch. 3); towards self-directed goals and skills* (hypothesised)